

● HARD

● ADVANCED MATH

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# Nonlinear functions

2 questions · ~3 min

10

**Q1**

ADVANCED MATH

The functions  $f$  and  $g$  are defined by the given equations, where  $x \geq 0$ . Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where  $x \geq 0$ ?

I.  $fx = 18(1.25)^x + 41$

II.  $gx = 90.73^x$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

A submersible device is used for ocean research. The function  $gx = -\frac{1}{55}x + 19x - 35$  gives the depth below the surface of the ocean, in meters, of the submersible device  $x$  minutes after collecting a sample, where  $x > 0$ . How many minutes after collecting the sample did it take for the submersible device to reach the surface of the ocean?

STUDENT-PRODUCED RESPONSE

|                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| .                    | /                    | .                    | .                    |
| 0                    | 0                    | 0                    | 0                    |
| 1                    | 1                    | 1                    | 1                    |
| 2                    | 2                    | 2                    | 2                    |
| 3                    | 3                    | 3                    | 3                    |
| 4                    | 4                    | 4                    | 4                    |
| 5                    | 5                    | 5                    | 5                    |
| 6                    | 6                    | 6                    | 6                    |
| 7                    | 7                    | 7                    | 7                    |
| 8                    | 8                    | 8                    | 8                    |
| 9                    | 9                    | 9                    | 9                    |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.